Robby Cochran

Staff Engineer / Engineering Manager - Runtime Security, Kubernetes, eBPF, Software Verification, AI-Agent Safety

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PROFILE

Security researcher and engineering leader with a CS PhD and publication record across NDSS, NSDI, POPL, and TISSEC. Builds cloud-native runtime security systems at the boundary of Kubernetes, eBPF observability, policy enforcement, behavioral detection, software verification, and AI-agent safety infrastructure. Experience spans early-stage startup engineering, Red Hat/IBM product integration, distributed-team leadership, large-enterprise design partners, technical promotion mentorship, and secure agent workflow strategy.

SELECTED IMPACT

- Built and led runtime-security product work for StackRox/RHACS across behavioral detection, process, network, and file activity, baselines, Kubernetes-native policy, and analyst/developer workflows.
- Helped carry StackRox from early startup product through Red Hat acquisition and into RHACS/OpenShift enterprise scale; product context included $0 to $50M+ revenue and 100+ releases.
- Led ACS agentic foundations: internal hackathon, CI tooling, support triage bots, skills strategy, secure harness primitives, and developer paths from local sandbox to OpenShift execution.
- Worked with large customers and design partners to translate deployment constraints into roadmap decisions around scale, compliance, performance, platform support, IBM Power/Z, and false-positive reduction.
- Current thesis: enterprise AI systems need runtime contracts, evaluation harnesses, observability, policy, and behavior-level security controls around agents - not just model-level safety.

EXPERIENCE

IBM / Red Hat — 2021 - Present
Staff Engineer / Engineering Manager, Red Hat Advanced Cluster Security (RHACS) — San Francisco, CA / Remote

- Led RHACS runtime-security roadmap spanning process activity, network security, file activity, behavioral baselines, runtime anomaly detection, Kubernetes-native policy workflows, and developer/analyst experience.
- Drove eBPF-based runtime visibility design for OpenShift/Kubernetes environments, including collection semantics, policy exceptions, performance constraints, platform support, and low false-positive detection goals.
- Built and guided secure AI-agent infrastructure through harness-openshell: declarative workflows, provider/credential boundaries, inference routing, sandbox execution, and repeatable CI validation.
- Managed distributed engineering teams while staying close to architecture and implementation; promoted L4 and L5 engineers through scope design, feedback, and career planning.
- Shaped OpenShift-first RHACS strategy around console integration, RBAC alignment, GitOps/policy-as-code, runtime observability, and AI-assisted workflows. Public talk: eBPF 101.

StackRox, Inc. (acquired by Red Hat) — 2017 - 2021
Staff Software Engineer / Senior Software Engineer / Member of Technical Staff — Mountain View, CA

- Joined an early-stage Kubernetes security startup and built runtime detection and enforcement features for containerized workloads and Kubernetes clusters.
- Advanced core runtime-security capabilities including baseline-driven detection, process and network visibility, and connections between observed runtime behavior and security policy.
- Worked across product, engineering, and customer-driven security cases with large design partners, helping mature the platform through acquisition by Red Hat.
- Patent: US 11,811,804, “System and method for detecting process anomalies in a distributed computation system utilizing containers.” Talk: BSidesSF 2018 - Listen to your Engine.

University of North Carolina at Chapel Hill — 2008 - 2016
PhD Researcher, Computer Security and Software Verification — Chapel Hill, NC

- Developed symbolic client verification: server-side techniques for checking whether observed client network traffic could have been produced by sanctioned software.
- Built symbolic-execution systems for online games, cryptographic protocols, and distributed applications; refactored KLEE internals for parallelism and performance research.
- Publications: NDSS 2010 Best Paper / TISSEC 2011 on online-game client verification; NDSS 2013 on distributed-application client behavior; NSDI 2017 on known cryptographic clients.

Microsoft Research - Research in Software Engineering (RiSE) — Summer 2013
Research Intern — Redmond, WA

- Program synthesis via crowdsourcing: combined symbolic automata, genetic programming, and online workers; publication: POPL 2015 “Program Boosting”; patent: US 9,753,696.

Intel Corporation - Visual Computing Group — 2006 - 2007
Software Engineer — Hillsboro, OR

- Built graphics-hardware analysis tooling to interpret, record, and replay Direct3D/OpenGL API calls for performance analysis of prototype visual-computing hardware.

EDUCATION

PhD, Computer Science, University of North Carolina at Chapel Hill, 2016 - Advisor: Michael K. Reiter. **BS, Computer Science**, Clemson University, Magna Cum Laude, 2006.

TECHNICAL DOMAINS

Runtime security; Kubernetes; OpenShift; RHACS; eBPF observability; behavioral baselines; policy-as-code; distributed systems; symbolic execution; software verification; formal methods; program synthesis; AI-agent safety; LLM gateways; MCP; Go; C/C++; Python; Rust; Bash; Docker; LLVM; SMT/SAT solvers.